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A LSB Steganography Detection Algorithm

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***Abstract*—The spatial LSB Steganography results in the alteration of the smooth characteristics between adjoining pixels of the raw image. The relation between the length of embedded message and the gradient energy is theoretically analyzed, and then a steganalysis and detection method, named Gradient Energy–Flipping Rate detection (GEFR) is proposed. Through the analysis of the variation of the gradient energy, which results from the LSB Steganography in color and grayscale image, the secret message embedded in the target image is detected, and the length of the embedded message is estimated. The method is proved effective and accurate by simulation (detection rate reaches 0.01bit per pixel).**

I. INTRODUCTION

Steganography is a new branch of secret communication [1-2]. While classical cryptography is about concealing the content of messages, steganography is about concealing their existence. We can use digital images, videos, sound files, and other computer files that contain perceptually irrelevant or redundant information as covers or carriers to hide secret messages. After embedding a secret message into the cover images, we obtain a so-called stego-image. It's important that the stego-images don't contain any detectable artifacts due to messages embedding. A third party could use such artifacts as an indication that a secret message is present. Once a third party can reliably identify which images contain secret messages, the steganographic tool becomes useless.

Obviously, the less information we embed into the cover image, the less probability of detecting artifacts. Another important factor is the choice of the cover image. Images with a low number of colors, computer art, and images with unique semantic content (such as fonts) should be avoided as cover images.

Fridrich et al. [3] showed that images stored previously in JPEG format are a poor choice for cover images. The quantization introduced by JPEG compression can serve as unique fingerprint. If you inspect the compatibility of the stego-image with the JPEG format, you can detect even small modifications of the cover image. But the steganalytic method also has at least two limitations. Firstly, it only applies to grayscale images that were originally stored in the JPEG format. And secondly, it can only be used offline for its computational complexity.

LSB embedding in palette images is a very popular steganography method. Johnson and Jajodia [4] pointed out that this method is very vulnerable. In order to swap the grayscale values of adjoining pixels, several steganographic programs create clusters of close palette colors. However, others can easily detect the suspicious pairs (clusters) of colors. Therefore, LSB hiding based on palette images is not safe enough. For the purpose of a highly safe convert communication, the only way is to use original and natural grayscale or colored images as covers.

Because in the original image, every pixel is directly represented by its color or grayscale value, the number of colors can't be reduced or limited for the image format. At the same time, the grayscale value's LSB presents random distribution. So it is difficult to analyze and detect this kind of target images. In this aspect, some valuable research results have been achieved [5-8]. Fridrich et al. [5] developed a steganographic method for detecting LSB embedding in24-bit color images—the Raw Quick Pairs (RQP) method. The new method is based on analyzing close pairs of colors created by LSB embedding. On the condition that the number of unique colors in the cover image is less than 30 percent that of the total pixels, it works reasonably well. When the number of unique colors exceeds about 50 percent that of total pixels, the results gradually become unreliable. This frequently happens for high resolution raw scans and images taken with digital cameras stored in an uncompressed format. Another disadvantage of the RQP method is that it can't be applied to grayscale images.

Fridrich and Goljan [6] proposed LSB Steganography dual-detection method, named RS method, based on probability statistics in the color or grayscale images. The basic idea is that LSB plane seems random in the typical cover images, but to some extent it could be predicted by the other 7 bit planes. Guo-Shiang Lin [7] proposed a feature analysis scheme based on gradient energy in the spatial domain and Laplacian parameters in the DCT domain. To detect the secret message, the original images are needed. Thus its application is limited.

In this paper, a novel blind detection method, named Gradient Energy–Flipping Rate detection (GEFR) is proposed. Through the analysis of the variation of the gradient energy that results from the spatial LSB embedding, the secret message is detected, and the length of the embedded message is estimated.

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II. GRADIENT ENERGY

A. Definition of Gradient Energy

Consider the 1-D data $I(n)$, we define the gradient $r(n)$ before data embedding to be

$$r(n) = I(n) - I(n-1) \tag{1}$$

Definition 1: The gradient Energy of $I(n)$ is defined and calculated as:

$$\mathrm{GE} = \sum |I(n) - I(n-1)|^2 = \sum |r(n)|^2 \tag{2}$$

After the secret data are inserted, $I(n)$ changes to be $I'(n)$ and the gradient is re-written as:

$$\begin{aligned} r'(n) &= I'(n) - I'(n-1) \\ &= (I(n) + S(n)) - (I(n-1) + S(n-1)) \\ &= r(n) + S(n) - S(n-1) \end{aligned} \tag{3}$$

Where $S(n) = I'(n) - I(n)$, represents the disturbing amount for $I(n)$ and is closely related to the secret data and the signal itself.

For spatial LSB embedding method, assuming that the embedding bits and the LSB plane of original image are essentially random, then we can obtain that

	$I'(n)=0$	$I'(n)=1$
$I(n)=0$	$S(n)=0$	$S(n)=1$
$I(n)=1$	$S(n)=-1$	$S(n)=0$

The pdf (probability distribution function) of $S(n)$ is:

$$p_{S(n)=0} = \frac{1}{2}; \quad p_{S(n)=\pm 1} = \frac{1}{4}$$

The gradient energy of the modified signal $I'(n)$ is then computed as

$$\begin{aligned} \mathrm{GE}' &= \sum |r'(n)|^2 = \sum |r(n) + S(n) - S(n-1)|^2 \\ &= \sum |r(n) + \Delta(n)|^2 \end{aligned} \tag{4}$$

Where $\Delta(n) = S(n) - S(n-1)$.

B. Relation between gradient energy and the length of embedding bits

Theorem 1: The variation of gradient energy introduced by LSB embedding only depends on embedding length, and the expectation of gradient energy variation equals to embedding length, i.e.

$$E\{GE' - GE\} = L \tag{5}$$

Proof:

$$\begin{aligned} E\{GE' - GE\} &= E\{\sum (r^2(n) + \Delta^2(n) + 2r(n)\Delta(n)) - \sum r^2(n)\} \\ &= E\{\sum r^2(n)\} + E\{\sum \Delta^2(n)\} + 2E\{\sum r(n)\Delta(n)\} - E\{\sum r^2(n)\} \\ &= E\{\sum \Delta^2(n)\} + 2E\{\sum r(n)\Delta(n)\} \end{aligned} \tag{6}$$

So there are two factors $E\{\sum \Delta^2(n)\}$ and $2E\{\sum r(n)\Delta(n)\}$ that influence the change of gradient energy.

(1) The influence of $E\{\sum \Delta^2(n)\}$

$\Delta(n)$ is uniformly distributed within [-2,+2], and thus

	$S(n)=0$	$S(n)=1$	$S(n)=-1$
$S(n-1)=0$	$\Delta(n)=0, p=\frac{1}{4}$	$\Delta(n)=1, p=\frac{1}{8}$	$\Delta(n)=-1, p=\frac{1}{8}$
$S(n-1)=1$	$\Delta(n)=-1, p=\frac{1}{8}$	$\Delta(n)=0, p=\frac{1}{16}$	$\Delta(n)=-2, p=\frac{1}{16}$
$S(n-1)=-1$	$\Delta(n)=1, p=\frac{1}{8}$	$\Delta(n)=2, p=\frac{1}{16}$	$\Delta(n)=0, p=\frac{1}{16}$

That is $p_{\Delta(n)=0} = \frac{3}{8}$, $p_{\Delta(n)=\pm 1} = \frac{1}{4}$, $p_{\Delta(n)=\pm 2} = \frac{1}{16}$

Assuming the length of embedding bits is L, and the pixel number of original image is M×N, then

$$E\{\sum_{l=1}^{L} \Delta^2(n)\} = \sum_{l=1}^{L} E\{\Delta^2(n)\} = L \tag{7}$$

(2) The influence of $E\{\sum r(n)\Delta(n)\}$

Assuming the independence between the embedding data and the original image, since the expectation of $\Delta(n)$ equals to 0, $E\{\sum r(n)\Delta(n)\} = 0$.

In summary, $E\{GE' - GE\} = L$, and the gradient energy variation introduced by LSB embedding only depends on the embedding length.

III. FLIPPING ALGORITHM

Algorithm 1: Flipping

We define an invertible operation F on set P called flipping. Flipping is a permutation of gray levels that entirely consists of 2-cycles, Thus $F(F(x)) = x$ for all $x \in P$. For an 8-bit grayscale image, $P=\{0, 1, ..., 255\}$. In this paper, the flipping of the LSB is used: $0 \leftrightarrow 1$, $2 \leftrightarrow 3$, $4 \leftrightarrow 5 \ldots$

The F operation holds the following three properties: If the cover image has $M \times N$ pixels and the embedding length is L ($L \le M \times N$),

1) For $L = M \times N$, there are $\frac{M \times N}{2}$ pixels with LSB flipped (negated), i.e. the flipping rate is 50%. The corresponding gradient energy is labeled as $\mathrm{GE}(\frac{M \times N}{2})$.

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2) The gradient energy of the original image is represented by GE(0). Flipping the LSBs of all pixels, the resulted gradient energy is labeled as GE(M×N). According to theorem 1, the gradient energy increment corresponding to *GE(M×N)* is twice that corresponding to $GE(\frac{M \times N}{2})$.

3) For $L<M \times N$, there are *L/2* pixels with LSB flipped (negated) (i.e. the flipping rate is $\frac{L/2}{M \times N}$). We denote the modified image as *I(L/2)*. The corresponding gradient energy is $GE(\frac{L/2}{M \times N}) = GE(0) + L$. Based on theorem 1, if we perform operation F to image *I(L/2)*, the resulted GE is $GE(\frac{M \times N - L/2}{M \times N})$, and

$$GE(\frac{L/2}{M \times N}) - GE(0) + GE(\frac{M \times N - L/2}{M \times N}) - GE(0) = 2 \times M \times N \quad (8)$$

In summary, we conclude that:

1) The gradient energy of the original image is $GE(0)$. Performing F to all LSBs, the resulted gradient energy is given by $GE(M \times N) = GE(0) + 2M \times N$.

2) For embedding length L=M×N, the modified image is noted as I(M × N/2), and the resulted gradient energy $GE(\frac{M \times N}{2}) = GE(0) + M \times N$. Performing F to image *I(M×N/2)*, the resulted gradient energy equals to $GE(\frac{M \times N}{2})$.

3) For L< M×N, the modified image is noted as I(L/2), the LSB flipping rate is $\frac{L/2}{M \times N}$ and $GE(\frac{L/2}{M \times N}) = GE(0) + L$. Performing *F* to image *I(L/2)*, the resulted $GE(\frac{M \times N - L/2}{M \times N}) = GE(0) + (2M \times N - L)$.

4) $GE(x)$ curve is modeled well with straight line $GE(x) = GE(0) + (2M \times N)x$, where x represents LSB flipping rate.

The gradient energies with various LSB flipping rates for 512×512 pixel images, namely LENA, BOAT, PEPPERS and ELANINE are calculated and shown in Fig. 1. From Fig. 1, we can see that (1) the gradient energy is linear with the flipping rate. (2) The gradient energy difference between the flipping rate 100% and 0% approximates 512×512. Thus theorem 1 is proved experimentally.

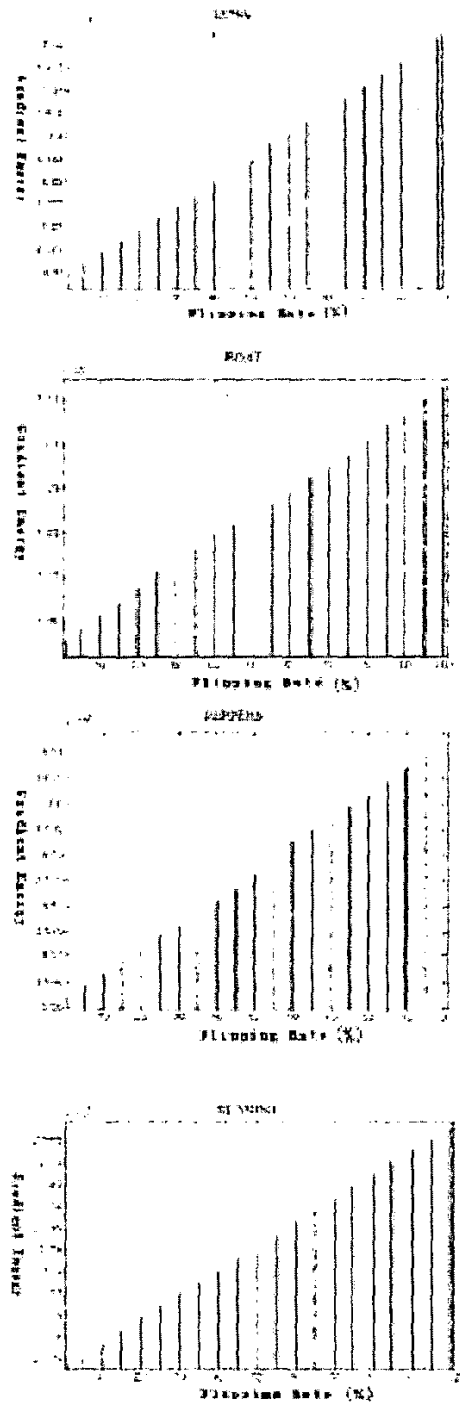

Fig.1 Relation between Gradient Energy and flipping rate for Lena, Boat, Peppers and Elaine

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IV. STEGANALYTIC METHOD PROPOSED AND THE EXPERIMENTAL RESULTS

Based on theorem 1 and the properties of flipping algorithm, a steganalytic method named "Gradient Energy–Flipping Rate detection (GEFR)" is proposed:

1) Calculating the gradient energy of the test image $GE(\frac{L/2}{M \times N})$;

2) Performing operation F to the test image and calculating $GE(\frac{M \times N - L/2}{M \times N})$;

3) Then calculating $GE(\frac{M \times N}{2}) = [GE(\frac{L/2}{M \times N}) + GE(\frac{M \times N - L/2}{M \times N})]/2$;

4) $GE(0)$ is thus given based on $GE(\frac{M \times N}{2}) = GE(0) + M \times N$

5) The estimated embedding length $L' = GE(\frac{L/2}{M \times N}) - GE(0)$.

To test the performance of Gradient Energy–Flipping Rate detection (GEFR) method, we used a large database of grayscale and color images. The simulation results for the four images mentioned in section 2 are shown in Fig. 2. We can see that: When embedding rate>0.05 bits per pixel, the method can reliably detect the presence of secret message and the embedding length estimation error is constraint within 10%. When embedding rate <0.01 bits per pixel, the presence detection becomes progressively unreliable.

Fig.2 Relation between embedding length estimation error rate and embedding rate. The four curves denote 512×512 Peppers, Boat, Lena, Elaine from above to below.

There are two reasons for the error reflected in the experiment:

1. Suppose the LSBs of the original images are randomly and evenly distributed.

2. Suppose the embedded secret messages had random and even distribution.

To improve the performance of GEFR method, we can calculate separately the variation of the horizontal and vertical gradient energy and get two estimated message length. Using arithmetical mean value of the above two estimated length, can reduce the influence of the non-random and uneven characteristics of raw image LSBs and embedded message.

V. CONCLUSIONS AND FUTURE WORK

The relation between gradient energy and LSB flipping rate is analyzed and a Gradient Energy–Flipping Rate detection (GEFR) method is proposed in this paper. When embedding rate>0.05 bits per pixel, the technique can reliably detect the presence of secret message and the embedding length estimation error is constraint within 10%. An upper bound of 0.01 bits per pixel is experimentally safe for spatial LSB embedding.

Our future research is focused on improving the embedding length estimation precision, especially when the embedding length is relatively short.

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